

1. Customer Feedback Validation System

A company collects customer ratings through a console application. Users sometimes enter text instead of numbers, leave the value blank, or enter ratings outside the allowed range (1–5).

Task:

Create a program that:

- Accepts a customer rating.
 - Ensures the rating is between 1 and 5.
 - Displays appropriate error messages for invalid inputs.
 - Continues asking until a valid rating is entered.
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2. Monthly Log Archive Manager

A software company stores daily system logs in a text file named **logs.txt**.

logs.txt

```
Server Started Successfully  
User Login: Aman  
Database Backup Completed  
User Logout: Aman  
Server Shutdown Initiated
```

Task:

Create a program that:

- Reads the file.
 - Displays the total number of lines.
 - Displays the total number of words.
 - Displays the first log entry.
 - Displays the last log entry.
 - Handles the situation when the file is missing.
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3. Warehouse Inventory Investigation

A warehouse manager suspects that some storage racks contain significantly more products than others.

Data:

```
inventory = [120, 85, 150, 60, 95, 180]
```

Task:

- Store the data in an appropriate numerical structure.
 - Find the total stock available.
 - Find the highest and lowest stock quantity.
 - Calculate the average stock.
 - Display all stock quantities greater than 100.
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4. Employee Attendance Investigation

An HR executive receives attendance records for ten consecutive days.

Data:

```
attendance = ["P", "P", "A", "P", "P", "A", "P", "P", "P", "A"]
```

Task:

- Display the first three records.
 - Display the last four records.
 - Display every alternate record.
 - Create a new list containing the attendance records repeated three times.
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5. Fuel Price Revision Notice

A fuel company decides to increase fuel prices by ₹3.50 per litre at all stations.

Data:

```
prices = [96.5, 97.2, 95.8, 98.1, 96.9]
```

Task:

- Increase all prices by ₹3.50 using a single operation.
 - Display old prices and revised prices.
 - Calculate the difference for each station.
 - Calculate the average revised fuel price.
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6. Software Asset Tracking Dashboard

An IT department maintains records of software installed on employee systems.

Data:

```
software = {  
    "Chrome": 45,
```

```
"VS Code": 32,  
"Python": 28,  
"MySQL": 20,  
"Git": 25  
}
```

Task:

Create a program that:

- Converts the software names and installation counts into a suitable tabular format.
 - Displays all software records.
 - Find the software with the highest installation count.
 - Find the total number of installations.
 - Display only software installed on more than 25 systems.
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7. Smart Factory Temperature Monitoring

Temperature sensors installed in different production units report hourly temperatures.

Data:

```
temperatures = [28.5, 30.2, 29.8, 31.1, 27.9, 30.5, 29.0]
```

Task:

- Store the temperature readings in a suitable analytical structure.
 - Calculate average temperature.
 - Find the highest and lowest temperatures.
 - Display temperatures above the average.
 - Generate a summary table containing all readings.
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8. Retail Store Sales Analysis

A retail company wants to analyze product sales.

Data:

```
sales_data = {  
    "Product": ["Laptop", "Mobile", "Printer", "Laptop", "Mobile"],  
    "Sales": [50000, 30000, 12000, 45000, 28000]  
}
```

Task:

- Create a tabular structure using the given data.
- Calculate total sales.
- Calculate average sales.

- Find the highest sales value.
 - Display all records where sales exceed ₹25,000.
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9. Employee Salary Register

An organization wants to maintain employee salary records.

Data:

```
names = ["Aman", "Riya", "Ali", "Sneha", "Rahul"]
salaries = [35000, 42000, 38000, 45000, 40000]
```

Task:

- Create a salary record structure.
 - Create a table containing employee names and salaries.
 - Display employees earning above ₹40,000.
 - Add a new column named **Bonus** containing 10% of salary.
 - Display the updated table.
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10. Production Quality Inspection Report

A manufacturing company stores product quality scores in a text file named **quality.txt**.

quality.txt

```
78
85
91
67
88
95
72
80
```

Task:

- Read all quality scores from the file.
- Handle file-related errors.
- Store the scores in a suitable numerical structure.
- Calculate average, maximum, and minimum score.
- Create a tabular report containing all scores.
- Display only the scores that are above the average.